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Papers 358 (concluded), 359, 360, 361, 362, 363 (begun)

This installment contains the concluding pages of Paper 358 ("Addition to the Memoir on Tschirnhausen's Transformation"), Paper 359 ("A Supplementary Memoir on Caustics"), Paper 360 ("Note on a Quartic Surface"), Paper 361 ("On Quartic Curves"), Paper 362 ("Note on Lobatschewsky's Imaginary Geometry"), and the opening of Paper 363 ("On the Theory of the Evolute").

Paper 358 (concluded): Addition to the Memoir on Tschirnhausen's Transformation

[*From the Philosophical Transactions, etc.*]

But I did not obtain an expression for the cubinvariant of the same function: such expression, it was remarked, would contain the square of the invariant Φ'; it was probable that there existed an identical equation,

$$JU'^3 - IU'^2H' + 4H'^3 + M\Theta' \;=\; -\Phi'^2,$$

which would serve to express Φ'^2 in terms of the other invariants; but, assuming that such an equation existed, the form of the factor M remained to be ascertained; and until this was done, the expression for the cubinvariant could not be obtained in its most simple form. I have recently verified the existence of the identical equation just referred to, and have obtained the expression for the factor M; and with the assistance of this identical equation I have obtained the expression for the cubinvariant of the form

$$(1,\, 0,\, \mathfrak{S},\, 2\mathfrak{T} \mid y,\, 1)^4.$$

The expression for the quadrinvariant was, as already mentioned, given in the former memoir: I find that the two invariants are in fact the invariants of a certain linear function of U, H; viz. the linear function is $= U'U + \frac{4}{3}\Theta'H$; so that, denoting by I^*, J^*, the quadrinvariant and the cubinvariant respectively of the form

$$(1,\, 0,\, \mathfrak{S},\, 2\mathfrak{T} \mid y,\, 1)^4,$$

we have

$$I^* = I(U'U + \tfrac{4}{3}\Theta'H),$$
$$J^* = J(U'U + \tfrac{4}{3}\Theta'H),$$

where I, J signify the functional operations of forming the two invariants respectively. The function $(1,\, 0,\, \mathfrak{S},\, 2\mathfrak{T} \mid y,\, 1)^4$, obtained by the application of Tschirnhausen's transformation to the equation

$$(a,\, b,\, c,\, d,\, e \mid x,\, 1)^4 = 0,$$

has thus the same invariants with the function

$$U'U + 4\Theta'H \;=\; U'\,(a,\, b,\, c,\, d,\, e \mid x,\, 1)^4 + 4\Theta'\,(ac - b^2,\; ad - bc,\; ae + 2bd - 3c^2,\; be - cd,\; ce - d^2 \mid x,\, 1)^4,$$

and it is consequently a linear transformation of the last-mentioned function; so that the application of Tschirnhausen's transformation to the equation $U = 0$ gives an equation linearly transformable into, and thus virtually equivalent to, the equation

$$U'U + 4\Theta'H = 0,$$

which is an equation involving the single parameter $\dfrac{4\Theta'}{U'}$: this appears to me a result of considerable interest. It is to be remarked that Tschirnhausen's transformation, wherein y is put equal to a rational and integral function of the order $n-1$ (if n be the order of the equation in x), is not really less general than the transformation wherein y is put equal to any rational function $\dfrac{V}{W}$ whatever of x; such rational function may, in fact, by means of the given equation in x, be reduced to a rational and integral function of the order $n-1$; hence in the present case, taking V, W to be respectively of the order $n-1$, $=3$, it follows that the equation in y obtained by the elimination of x from the equations

$$(a,\, b,\, c,\, d,\, e \mid x,\, 1)^4 = 0,$$

$$y \;=\; \frac{(\alpha,\, \beta,\, \gamma,\, \delta \mid x,\, 1)^3}{(\alpha',\, \beta',\, \gamma',\, \delta' \mid x,\, 1)^3}$$

is a mere linear transformation of the equation $AU + BH = 0$, where A, B are functions (not as yet calculated) of $(a, b, c, d, e, \alpha, \beta, \gamma, \delta, \alpha', \beta', \gamma', \delta')$.

Article Nos. 1, 2, 3. Investigation of the identical equation

$$JU'^3 - IU'^2H' + 4H'^3 + M\Theta' \;=\; -\Phi'^2.$$

1. It is only necessary to show that we have such an equation, M being an invariant, in the particular case $a = e = 1$, $b = d = 0$, $c = \theta$, that is for the quartic function $(1,\, 0,\, \theta,\, 0,\, 1 \mid x,\, y)^4$; for, this being so, the equation will be true in general. Writing the equation in the form

$$-M\Theta' \;=\; U'^2(JU' - IH') + 4H'^3 + \Phi'^2,$$

and observing that we have

$$\begin{gathered} U'^2 = (B^2 + D^2) + 2\theta BD + 4\theta^2C^2, \\ H' = \theta(B^2 + D^2) + (1 + \theta^2)BD - 4\theta^2C^2, \\ \Theta' = BD - C^2, \\ \Phi' = (1 - 9\theta^2)\,C\,(B^2 - D^2), \\ I = 1 + 3\theta^4, \\ J = \theta - \theta^5, \end{gathered}$$

and thence

$$JU' - IH' \;=\; -4\theta^3(B^2 + D^2) + (-1 - 2\theta^2 - 5\theta^4)BD + (8\theta + 8\theta^3)C^2,$$

the equation becomes

$$\begin{aligned} -(BD - C^2)\,M \;=\; &\big\{-4\theta^3(B^2 + D^2) + (-1 - 2\theta^2 - 5\theta^4)BD + (8\theta + 8\theta^3)C^2\big\} \\ &\times \big\{(B^2 + D^2) + 2\theta BD + 4\theta^2C^2\big\} \\ &+ 4\big[\theta(B^2 + D^2) + (1 + \theta^2)BD - 4\theta^2C^2\big]^3 \\ &+ (1 - 9\theta^2)^2\,C^2\,(B^2 - D^2)^2. \end{aligned}$$

2. It is found by developing that the right-hand side is in fact divisible by $BD - C^2$, and that the quotient is

$$\begin{aligned} = &(-1 + 10\theta^2 - 9\theta^4)(B^2 + D^2)^2 \\ &+ (8\theta + 16\theta^3 - 24\theta^5)(B^2 + D^2)BD \\ &+ (4 + 8\theta^2 + 4\theta^4 - 16\theta^6)B^2D^2 \\ &+ (-64\theta^2 - 192\theta^6)(B^2 + D^2)C^2 \\ &+ (16\theta - 416\theta^3 - 112\theta^5)BDC^2 \\ &+ (-128\theta^2 + 128\theta^6)C^4. \end{aligned}$$

3. This is found to be

$$\begin{aligned} &= -I^2U'^2 + 12JU'H' + 4IH'^2 \\ &\quad - 168JU'^2\Theta' \cdot \theta^3 \\ &\quad - (\text{further terms involving } \Theta' \text{ etc.}) \end{aligned}$$

which is consequently the value of $-M$. We have therefore

$$\begin{aligned} -\Phi'^2 &= JU'^3 - IU'^2H' + 4H'^3 \\ &\quad + U'^2\big[I^2 - 12JU'H' - 4IH'^2\big]\Theta' \\ &\quad + 168JU'^2\,\Theta'\,\theta^3 + \cdots, \end{aligned}$$

which is the required identical equation.

Article No. 4. Calculation of the Cubinvariant.

4. We have

$$J^* \;=\; -\big((HH' - \tfrac{1}{12}IU'^2)\Theta'\big)\big[IH'^2 - 3H'^3 + (12JU' + 2IH')\Theta' + I^2\Theta'^2\big] - \Phi'^2,$$

whence, substituting for $-\Phi'^2$ its value and reducing, we find

$$J^* \;=\; JU'^3 + \Theta' \cdot \tfrac{3}{2}I^2U'^2 + \Theta'^2(4IJU') + \Theta'^3\big(16J^2 - \tfrac{16}{27}I^3\big).$$

Article No. 5. Final expressions of the two Invariants.

The value of I^* has been already mentioned to be

$$I^* \;=\; IU'^2 + \Theta' \cdot 12JU' + \Theta'^2 \cdot \tfrac{2}{3}I^2,$$

and it hence appears that the values of the two invariants may be written

$$I^* = (I,\, 18J,\, 3I^2 \mid U',\, \tfrac{2}{3}\Theta')^2,$$

$$J^* = (J,\, I^2,\, 9IJ,\, -I^3 + 54J^2 \mid U',\, \tfrac{2}{3}\Theta')^3.$$

But we have (see Table No. 72 in my "Seventh Memoir on Quantics," *Philosophical Transactions*, vol. CLI. (1861), pp. 277–292, [269])

$$I(\alpha U + 6\beta H) \;=\; (I,\, 18J,\, 3I^2 \mid \alpha,\, \beta)^2,$$

$$J(\alpha U + 6\beta H) \;=\; (J,\, I^2,\, 9IJ,\, -I^3 + 54J^2 \mid \alpha,\, \beta)^3;$$

so that, writing $\alpha = U'$, $\beta = \frac{2}{9}\Theta'$, we have

$$I^* = I(U'U + \tfrac{4}{3}\Theta'H),$$

$$J^* = J(U'U + \tfrac{4}{3}\Theta'H);$$

or the function $(1,\, 0,\, \mathfrak{S},\, 2\mathfrak{T} \mid y,\, 1)^4$ obtained from Tschirnhausen's transformation of the equation $U = 0$ has the same invariants with the function $U'U + \frac{4}{3}\Theta'H$; or, what is the same thing, the equation $(1,\, 0,\, \mathfrak{S},\, 2\mathfrak{T} \mid y,\, 1)^4 = 0$ is a mere linear transformation of the equation $U'U + 4\Theta'H = 0$; which is the above-mentioned theorem.

Paper 359: A Supplementary Memoir on Caustics

[*From the Philosophical Transactions of the Royal Society of London*, vol. CLVII. (for the year 1867), pp. 7–16. Received November 15,—Read November 22, 1866.]

It is near the conclusion of my "Memoir on Caustics," *Philosophical Transactions*, vol. CXLVII. (1857), pp. 273–312, [145], remarked that for the case of parallel rays refracted at a circle, the ordinary construction for the secondary caustic cannot be made use of (the entire curve would in fact pass off to an infinite distance), and that the simplest course is to measure off the distance GQ from a line through the centre of the refracting circle perpendicular to the direction of the incident rays. The particular secondary caustic, or orthogonal trajectory of the refracted rays, obtained on the above supposition was shown to be a curve of the order 8; and it was further shown (by consideration of the case wherein the distance GQ is measured off from an arbitrary line perpendicular to the incident rays) that the general secondary caustic or orthogonal trajectory of the refracted rays was a curve of the same order 8. The last-mentioned curve in the case of reflexion, or for $\mu = -1$, degenerates into a curve of the order 6; and I propose in the present supplementary memoir to discuss this sextic curve, viz. the sextic curve which is the general secondary caustic or orthogonal trajectory of parallel rays reflected at a circle.

1. For parallel rays refracted at a circle, taking the equation of the circle to be $x^2+y^2=1$, and the incident rays to be parallel to the axis of x, then if $x=m$ be an arbitrary line perpendicular to the direction of the incident rays, the secondary caustic is the envelope of the circle

$$\mu^2\left[(x-\alpha)^2+(y-\beta)^2\right]-(x-m)^2 \;=\; 0,$$

where (α,β) are the coordinates of a variable point on the refracting circle, and as such satisfy the equation $\alpha^2+\beta^2=1$. Or, what is the same thing, writing $\alpha=\cos\theta$, $\beta=\sin\theta$, the secondary caustic is the envelope of the circle

$$\mu^2\left[(x-\cos\theta)^2+(y-\sin\theta)^2\right]-(x-m)^2 \;=\; 0,$$

where θ is a variable parameter.

2. The equation may be written

$$A\cos 2\theta+B\sin 2\theta+C\cos\theta+D\sin\theta+E \;=\; 0,$$

where

$$A=1,\quad B=0,$$
$$C=4\mu^2x-4m,\qquad D=4\mu^2y,$$
$$E=-2\mu^2(x^2+y^2)-2\mu^2+1+2m^2,$$

and which in the case of reflexion, or for $\mu=-1$, become

$$A=1,\quad B=0,$$
$$C=4x-4m,\qquad D=4y,$$
$$E=-2(x^2+y^2)-1+2m^2;$$

viz. the equation of the variable circle is in this case

$$\cos 2\theta+4(x-m)\cos\theta+4y\sin\theta+2m^2-1-2(x^2+y^2) \;=\; 0.$$

3. Now in general for the equation

$$A\cos 2\theta+B\sin 2\theta+C\cos\theta+D\sin\theta+E \;=\; 0,$$

where the coefficients are any functions whatever of the coordinates (x,y), the equation of the envelope is $S^3-T^2=0$, where

$$S \;=\; 12(A^2+B^2)-3(C^2+D^2)+4E^2,$$
$$-T \;=\; 27A(C^2-D^2)+54BCD-\big(72(A^2+B^2)+9(C^2+D^2)\big)E+8E^3.$$

4. Hence, substituting for A, B, C, D, E the above reflexion values, we find

$$S = 12 - 48\big((x-m)^2 + y^2\big) + 4\big(2m^2 - 1 - 2x^2 - 2y^2\big)^2,$$

$$\begin{aligned} -T &= 432\big((x-m)^2 - y^2\big) \\ &\quad - 72\Big(12 + 144\big((x-m)^2 + y^2\big)\Big)\big(2m^2 - 1 - 2x^2 - 2y^2\big) \\ &\quad + 8\big(2m^2 - 1 - 2x^2 - 2y^2\big)^3. \end{aligned}$$

Writing in these equations

$$\begin{aligned} (x-m)^2 + y^2 &= x^2 + y^2 - 2mx + m^2, \\ (x-m)^2 - y^2 &= 2x^2 - 2mx + m^2 - (x^2 + y^2), \end{aligned}$$

then after some simple reductions, we find

$$S = 16\big\{(x^2 + y^2 - m^2 - 1)^2 + 6m(x-m)\big\},$$

$$T = 32\big\{2(x^2 + y^2 - m^2 - 1)^3 + 18m(x-m)(x^2 + y^2 - m^2 - 1) - 27(x-m)^3\big\},$$

and thence

$$S^3 - T^2 = 1024\,(x-m)^2\,U,$$

where

$$\begin{aligned} U &= 4(x^2 + y^2 - m^2 - 1)^3 \\ &\quad + 4m^2(x^2 + y^2 - m^2 - 1)^2 \\ &\quad + 36m\,(x^2 + y^2 - m^2 - 1)(x-m) \\ &\quad - 27(x-m)^2 \\ &\quad + 32m^2(x-m), \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} U &= 4(x^2 + y^2)^3 \\ &\quad - (8m^2 + 12)(x^2 + y^2)^2 \\ &\quad + (36mx + 4m^4 - 20m^2 + 12)(x^2 + y^2) \\ &\quad - 27x^2 + (-4m^3 + 18m)x + m^2 - 4; \end{aligned}$$

so that the equation of the secondary caustic is $U = 0$, or the secondary caustic is, as stated above, a sextic curve.

5. It is easy to see that the foregoing envelope may be geometrically constructed as follows: viz. if from the point Q (coordinates $\cos\theta$, $\sin\theta$) on the reflecting circle we draw QM perpendicular to the line $x - m = 0$, and then from the point M draw MN perpendicular to QT, the tangent at T, and produce MN to a point P such that $PN = NM$, then P is a point of the envelope; and we thence obtain for the coordinates (x, y) of a point P of the envelope the values

$$x = m - 2(m - \cos\theta)\cos^2\theta,$$

$$y = \sin\theta - 2(m - \cos\theta)\cos\theta\sin\theta,$$

or, what is the same thing,

$$\begin{aligned} x &= 2\cos^3\theta - m(2\cos^2\theta - 1), \\ y &= \sin\theta(2\cos^2\theta + 1) - 2m\sin\theta\cos\theta, \end{aligned}$$

or, as these equations may also be written,

$$x = \tfrac{3}{2}\cos\theta - m\cos 2\theta + \tfrac{1}{2}\cos 3\theta,$$

$$y = \tfrac{3}{2}\sin\theta - m\sin 2\theta + \tfrac{1}{2}\sin 3\theta.$$

6. This result may be verified by showing that these values satisfy the equation

$$\cos 2\theta + 4(x-m)\cos\theta + 4y\sin\theta + 2m^2 - 1 - 2(x^2+y^2) = 0,$$

and also the derived equation

$$\sin 2\theta + 2(x-m)\sin\theta - 2y\cos\theta = 0.$$

We in fact have

$$x\sin\theta - y\cos\theta = m\sin\theta - \tfrac{1}{2}\sin 2\theta,$$

and thence

$$x\cos\theta + y\sin\theta = \tfrac{3}{2} - m\cos\theta + \tfrac{1}{2}\cos 2\theta,$$
$$(x-m)\sin\theta - y\cos\theta = -\tfrac{1}{2}\sin 2\theta,$$

which is one of the equations to be verified; and also

$$(x-m)\cos\theta + y\sin\theta = \tfrac{3}{2} - 2m\cos\theta + \tfrac{1}{2}\cos 2\theta.$$

We have moreover

$$x^2+y^2 = \tfrac{5}{2} + m^2 - 4m\cos\theta + \tfrac{3}{2}\cos 2\theta;$$

and, by means of these last equations, the other equation

$$\cos 2\theta + 4(x-m)\cos\theta + 4y\sin\theta + 2m^2 - 1 - 2(x^2+y^2) = 0,$$

is also verified.

7. The foregoing values of (x, y) give

$$dx = \left(-\tfrac{3}{2}\sin\theta + 2m\sin 2\theta - \tfrac{3}{2}\sin 3\theta\right)d\theta = -\sin 2\theta\,(3\cos\theta - 2m)\,d\theta,$$
$$dy = \left(\tfrac{3}{2}\cos\theta - 2m\cos 2\theta + \tfrac{3}{2}\cos 3\theta\right)d\theta = \cos 2\theta\,(3\cos\theta - 2m)\,d\theta,$$

or, what is the same thing, $dx : dy = -\sin 2\theta : \cos 2\theta$.

Hence taking for a moment (X, Y) as the current coordinates of a point in the tangent of the envelope, the equation of the tangent of the envelope is

$$X\,dy - Y\,dx = x\,dy - y\,dx,$$

or, substituting for x, y, dx, dy their values, this equation takes the very simple form

$$X\cos 2\theta - Y\sin 2\theta - 2\cos\theta + m = 0,$$

or writing (x, y) in place of (X, Y), that is taking now (x, y) as the current coordinates of a point in the tangent, the equation of the tangent is

$$x\cos 2\theta - y\sin 2\theta - 2\cos\theta + m = 0;$$

whence observing that this equation may be expressed as a rational equation of the fourth order in terms of the parameter $\tan\frac{1}{2}\theta$ (or $\cos\theta + \sqrt{-1}\sin\theta$), it appears that the class of the secondary caustic is $= 4$.

8. The secondary caustic may be considered as the envelope of the tangent, and the equation be obtained in this manner. Comparing with the general equation

$$A\cos 2\theta + B\sin 2\theta + C\cos\theta + D\sin\theta + E = 0,$$

we have

$$A = x, \quad B = -y, \quad C = -2, \quad D = 0, \quad E = m.$$
$$S = 4\left\{3(x^2+y^2) + m^2 - 3\right\},$$

giving

$$-T = 4\{18m(x^2+y^2) - 27x - 2m^3 + 9m\},$$
$$S^3 - T^2 = 16V,$$

if for a moment

$$V = 4\{3(x^2+y^2) + m^2 - 3\}^3 - \{18m(x^2+y^2) - 27x - 2m^3 + 9m\}^2.$$

The equation of the curve is thus obtained in the form $V = 0$; this should of course be equivalent to the before-mentioned equation $U = 0$; and by developing V, and comparing with the second of the two expressions of U, it appears that we in fact have $V = 27U$.

9. Taking as parameter $\tan\frac{1}{2}\theta$, or if we please $\cos\theta + \sqrt{-1}\sin\theta$, the foregoing values of (x, y) in terms of θ give $(x, y, 1)$ proportional to rational and integral functions of the degree 6 in the parameter; so that not only the curve is a sextic curve, but it is a unicursal sextic, or curve of the order 6 with the maximum number, $= 10$, of nodes and cusps; that is, if δ be the number of nodes and κ the number of cusps, we have $\delta + \kappa = 10$. Moreover, introducing the same parameter into the equation of the tangent, this equation is seen to be of the degree 4 in the parameter; that is, the class of the curve is $= 4$: this implies $2\delta + 3\kappa = 26$, and we have therefore $\delta = 4$, $\kappa = 6$. To verify these numbers, it is to be remarked that it appears by the equation of the curve that there is at each of the circular points at infinity a triple point in the nature of the point $x = 0$, $y = 0$ on the curve $y^3 = x^4$; such a point is in fact equivalent to a node and two cusps, and we have thus the two circular points at infinity counting together as 2 nodes and 4 cusps; there should therefore besides be 2 nodes and 2 cusps, and I proceed to establish the existence of these by means of the expressions for (x, y) in terms of θ.

10. To find the cusps, we have

$$\frac{dx}{d\theta} = -\sin 2\theta\,(3\cos\theta - 2m) = 0,$$

$$\frac{dy}{d\theta} = \cos 2\theta\,(3\cos\theta - 2m) = 0,$$

which are each of them satisfied if only $3\cos\theta - 2m = 0$, or $\cos\theta = \frac{2}{3}m$; the corresponding values of (x, y) are found to be

$$x = m - \tfrac{4}{27}m^3, \qquad y = \pm\big(1 - \tfrac{4}{9}m^2\big)^{3/2},$$

and we have thus two cusps situate symmetrically in regard to the axis of x; the cusps are real if $m < \frac{3}{2}$, imaginary if $m > \frac{3}{2}$; for $m = \frac{3}{2}$, the two cusps unite together at the point $x = \frac{1}{2}$ on the axis of x, giving rise to a higher singularity, which will be further examined, post. No. 12.

11. The curve is symmetrical in regard to the axis of x, and hence any intersection with the axis of x, not being a point where the curve cuts the axis at right angles, will be a node. Hence, in order to find the nodes, writing $y = 0$, this is

$$\sin\theta\,(1 - 2m\cos\theta + 2\cos^2\theta) = 0,$$

giving $\sin\theta = 0$, that is,

$$\theta = 0, \quad x = 2 - m;$$
$$\theta = \pi, \quad x = -2 - m;$$

but these are each of them ordinary points on the axis of x; or else giving

$$1 - 2m\cos\theta + 2\cos^2\theta = 0,$$

that is

$$\cos\theta = \tfrac{1}{2}\big(m \pm \sqrt{m^2 - 2}\big).$$

The corresponding values of x are

$$x = \cos\theta\,(2\cos^2\theta - 2m\cos\theta) + m = m - \cos\theta = \tfrac{1}{2}\big(m \mp \sqrt{m^2 - 2}\big);$$

each of the points in question, viz. the points

$$x \;=\; \tfrac{1}{2}\big(m \pm \sqrt{m^2-2}\big), \qquad y=0,$$

is a node on the axis of x.

12. It is to be observed that for $m < \sqrt{2}$ the nodes are both imaginary; for $m=\sqrt{2}$ they coincide together at the point $x=\dfrac{1}{\sqrt{2}}$; for $m>\sqrt{2}$ they are both real: it is to be further noticed that

$$\text{node,} \quad x=\tfrac{1}{2}\big(m+\sqrt{m^2-2}\big), \quad \text{corresponds to} \quad \cos\theta=\tfrac{1}{2}\big(m-\sqrt{m^2-2}\big),$$

where (m being $>\sqrt{2}$) the point $(\cos\theta, \sin\theta)$ is a real point on the circle $x^2+y^2=1$; in fact for $m<\frac{3}{2}$ (that is, $m=\sqrt{2}$ to $m=\frac{3}{2}$) we have $\frac{1}{2}(m-\sqrt{m^2-2}) < \frac{2}{3}m$, that is, $\cos\theta < \frac{2}{3}$; but $m=$ or $>\frac{3}{2}$, then $\cos\theta = \frac{1}{2}(m-\sqrt{m^2-2}) = \dfrac{1}{m+\sqrt{m^2-2}}$ is $=$ or $<\frac{2}{3}$, and

$$\text{node,} \quad x=\tfrac{1}{2}\big(m-\sqrt{m^2-2}\big), \quad \text{corresponds to} \quad \cos\theta=\tfrac{1}{2}\big(m+\sqrt{m^2-2}\big),$$

where (m being $>\sqrt{2}$) the point $(\cos\theta,\sin\theta)$ is a real point on the circle $x^2+y^2=1$ so long as m is not $>\frac{3}{2}$, that is, from $m=\sqrt{2}$ to $m=\frac{3}{2}$; but if $m>\frac{3}{2}$, then the point in question is an imaginary point on the circle—whence also the node $x=\frac{1}{2}(m-\sqrt{m^2-2})$ is an acnode or isolated point.

In the case $m=\frac{3}{2}$ we have

$$\text{node,} \quad x=\tfrac{1}{2}, \quad \text{corresponding to} \quad \cos\theta=\tfrac{1}{2} \text{ or } \theta=60^\circ,$$

$$\text{node,} \quad x=\tfrac{1}{2}, \quad \text{corresponding to} \quad \cos\theta=1 \text{ or } \theta=0^\circ,$$

the last-mentioned point $x=\frac{1}{2}$ being in fact the point of union of two cusps in the case $m=\frac{3}{2}$ now in question. Hence in this case we have at $(x=\frac{1}{2}, y=0)$ a triple point equivalent to two cusps and a node; visibly, there is only a single branch cutting the axis of x at right angles.

In the case $m=\sqrt{2}$, the nodes coincide as above mentioned at the point $x=\dfrac{1}{\sqrt{2}}$ on the axis; for this value of m the coordinates of the cusps are

$$x=\tfrac{7}{27}\sqrt{2}\ \big(=\tfrac{7}{27}\sqrt{2}, \text{ which is } <1+\sqrt{2}\big); \qquad y=\pm\tfrac{1}{27}.$$

13. Starting from the equation $1024(x-m)^2U=S^3-T^2=0$, it is clear that the cusps are included among the intersections of the curves $S=0$, $T=0$: these two curves intersect in 24 points which lie $9+9$ at the circular points at infinity, $2+2$ at the points $x=m$, $y^2-1=0$, and $1+1$ are the cusps, or points $x=m-\frac{4}{27}m^3$, $y^2=(1-\frac{4}{9}m^2)^3$. To verify this, writing for a moment

$$S' \;=\; (x^2+y^2-m^2-1)^2+6m(x-m),$$

$$T' \;=\; 2(x^2+y^2-m^2-1)^3+18m(x-m)(x^2+y^2-m^2-1)-27(x-m)^3,$$

then we have

$$\begin{aligned} T'-2(x^2+y^2-m^2-1)S' &= 6m(x-m)(x^2+y^2-m^2-1)-27(x-m)^3 \\ &= 3(x-m)\big\{2m(x^2+y^2-m^2-1)-9(x-m)^2\big\}; \end{aligned}$$

so that the equations $S=0$, $T=0$, or, what is the same thing, $S'=0$, $T'=0$ give

$$(x-m)\big\{2m(x^2+y^2-m^2-1)-9(x-m)^2\big\} \;=\; 0,$$

that is, $x-m=0$, or else $x^2+y^2-m^2-1=\dfrac{9}{2m}(x-m)^2$. And combining herewith the equation $S'=(x^2+y^2-m^2-1)^2+6m(x-m)=0$, we have $x-m=0$, $(y^2-1)^2=0$, or else

$$(x^2+y^2-m^2-1)^2 \;=\; \frac{81}{4m^2}(x-m)^4 \;=\; -6m(x-m),$$

and therefore

$$(x-m)\{27(x-m)-8m^3\} \;=\; 0,$$

the second factor of which gives $x = m - \frac{4}{27}m^3$, and thence $x^2+y^2-m^2-1 = \frac{4}{9}m^2$, that is, $x^2+y^2 = 1+\frac{1}{9}m^2$, and therefore $y^2 = (1-\frac{4}{9}m^2) - (m-\frac{4}{27}m^3)^2 = (1-\frac{4}{9}m^2)^3$, that is, we have

$$x = m - \tfrac{4}{27}m^3, \qquad y^2 = \big(1-\tfrac{4}{9}m^2\big)^3,$$

which, as appears above, gives the two cusps.

14. Similarly, in the equation $16\,V = S^3 - T^2 = 0$, the intersections of the curves $S = 0$, $T = 0$ must include the cusps; the curves in question are the two circles

$$3(x^2+y^2) + m^2 - 3 \;=\; 0,$$

$$18m(x^2+y^2) - 27x - 2m^3 + 9m \;=\; 0,$$

meeting in the circular points at infinity, and in the two cusps. It is to be added that the tangent at the cusps coincides with the tangent of the last-mentioned circle,

$$18m(x^2+y^2) - 27x - 2m^3 + 9m \;=\; 0,$$

or, as this may also be written,

$$\left(x - \tfrac{3}{4m}\right)^2 + y^2 \;=\; \left(\tfrac{4m^2-9}{12m}\right)^2.$$

15. The axis of x meets the secondary caustic in the two nodes counting as 4 intersections, and besides in 2 points, viz. the points $x = 2-m$, $x = -2-m$; these correspond to the values $\theta = 0$ and $\theta = \pi$ respectively. But to verify them by means of the equation of the curve, it may be remarked that for $y = 0$ we have

$$S = 4(3x^2+m^2-3), \qquad T = 4(18mx^2 - 27x - 2m^3 + 9m);$$

and writing herein $x = \pm 2 - m$, we find

$$S = 4(2m \mp 3)^2, \qquad T = 8(2m \mp 3)^3,$$

values which satisfy the equation $S^3 - T^2 = 0$.

16. In the equation $U = 0$ of the curve, writing $x - m = 0$, the equation becomes

$$4(y^2-1)^2 + 4m^2(y^2-1)^2\cdot 0 \;=\; 0,$$

that is

$$4(y^2-1)^2\,(y^2-1+m^2) \;=\; 0,$$

and the line $(x-m) = 0$ is thus a double tangent to the curve, touching it at the points $x = m$, $y = \pm 1$, and besides meeting it at the points $x = m$, $y = \pm\sqrt{1-m^2}$, that is, at the intersections of the line $x - m = 0$, with the circle $x^2+y^2 = 1$.

17. The maximum or minimum values of y correspond to the values $\theta = \frac{1}{4}\pi$, $\theta = \frac{3}{4}\pi$, $\theta = \frac{5}{4}\pi$, $\theta = \frac{7}{4}\pi$ of θ; and we have for

$$\begin{array}{lll} \theta = \frac{1}{4}\pi, & x = \frac{1}{2}\sqrt{2}, & y = \sqrt{2} - m, \\ \theta = \frac{3}{4}\pi, & x = -\frac{1}{2}\sqrt{2}, & y = \sqrt{2} + m, \\ \theta = \frac{5}{4}\pi, & x = -\frac{1}{2}\sqrt{2}, & y = -\sqrt{2} - m, \\ \theta = \frac{7}{4}\pi, & x = \frac{1}{2}\sqrt{2}, & y = -\sqrt{2} + m. \end{array}$$

18. It is now easy to trace the secondary caustic; we may without loss of generality assume that m is positive, and the values to be considered are

$$m = 0, \quad m = 1, \quad m = \sqrt{2}, \quad m = \tfrac{3}{2},$$

with the intermediate values $m > 0 < 1$, &c. ... and $m > \frac{3}{2}$. I have for convenience delineated in the figure only a portion of each curve, viz. the figure is terminated at the negative value $x = -\frac{1}{2}\sqrt{2}$, which corresponds to the maximum value $y = \sqrt{2} + m$; as x increases negatively, the value of the ordinate y diminishes continuously from this maximum value, becoming $= 0$ for the value $x = -2 - m$, and the curve at this point cutting the axis of x at right angles; this is a sufficient explanation of the form of the curves beyond the limits of the figure. Moreover the curve is symmetrical in regard to the axis of x, and I have within the limits of the figure delineated only one of the two halves of the curve.

19. For $m > \frac{3}{2}$ the cusps are both imaginary, the nodes both real, but one of them is an isolated point or acnode (shown in the figure by a small cross). The curve has an interior loop, as shown in the figure, and there is also the acnode lying within the loop.

For $m = \frac{3}{2}$, there is still an interior loop, but the acnode has united itself to the loop, the point of union, although presenting no visible singularity, being really a triple point equivalent to a node and two cusps. And in all the cases which follow there are two real cusps.

For $m =< \frac{3}{2} > \sqrt{2}$, the loop has altered its form in such wise as to exhibit the node and two cusps, the curve has therefore two real nodes.

[Figure: composite plot of the secondary caustic curves for the various values of m enumerated above; the figure (PDF p. 489) shows the half of the curve delineated above, with axes, the reflecting circle $x^2+y^2 = 1$, the line $x = m$, and the loop, cusps, nodes and acnode marked.]

For $m = \sqrt{2}$, the two nodes unite together into a tacnode, so that the loop is on the point of disappearing; and for $m < \sqrt{2} > 1$ the nodes are imaginary, and there is thus no longer any loop.

In all the above forms the double tangent $x = m$ touches the curve at the points $y = \pm 1$, but the other two intersections of the double tangent with the curve are imaginary.

For $m = 1$, the double tangent has the two coincident real intersections $y = 0$, or it is in fact a triple tangent.

For $m < 1 > 0$, the double tangent has with the curve two real intersections, viz. they are the points where the double tangent meets the circle $x^2 + y^2 = 1$.

And finally, for $m = 0$, the points in question unite themselves with the points of contact, the double tangent $x = 0$ being in this case the common tangent at the two cusps $x = 0$, $y = \pm 1$.

Added May 13, 1867.

20. As remarked in the original memoir, p. 312, the secondary caustic, in the last-mentioned case $m = 0$, is a curve similar to and double the magnitude of the caustic itself (viz. the caustic for parallel rays reflected at a circle), the position of the two curves differing by a right angle.

The secondary caustics corresponding to the different values of m form, it is clear, a system of parallel curves; and, by the remark just referred to, it appears that this system is similar to the system of curves parallel to the caustic for parallel rays reflected at a circle.

Paper 360: Note on a Quartic Surface

[*From the Philosophical Magazine*, vol. XXIX. (1865), pp. 19–22.]

It would, I think, be worth while to study in detail the quartic surface which is the envelope of a sphere having its centre on a given conic, and passing through a given point. The equations of the conic being $z = 0$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the coordinates of a point on the conic may be taken to be $x = a\cos\theta$, $y = b\sin\theta$, $z = 0$, whence, if (α, β, γ) be the coordinates of the given point, the equation of the sphere is

$$(x - a\cos\theta)^2 + (y - b\sin\theta)^2 + z^2 \;=\; (\alpha - a\cos\theta)^2 + (\beta - b\sin\theta)^2 + \gamma^2,$$

or, what is the same thing,

$$x^2 + y^2 + z^2 - \alpha^2 - \beta^2 - \gamma^2 - 2(x - \alpha)a\cos\theta - 2(y - \beta)b\sin\theta \;=\; 0;$$

and hence the equation of the surface is at once seen to be

$$\left(x^2 + y^2 + z^2 - \alpha^2 - \beta^2 - \gamma^2\right)^2 \;=\; 4a^2(x - \alpha)^2 + 4b^2(y - \beta)^2.$$

If $a = b$ (that is, if the conic be a circle), then we may without loss of generality write $\beta = 0$, and the equation then is

$$\left(x^2 + y^2 + z^2 - \alpha^2 - \gamma^2\right)^2 \;=\; 4a^2\left[(x-\alpha)^2 + y^2\right].$$

This may be written

$$\left(x^2 + y^2 + z^2 - \alpha^2 - \gamma^2 - 2a^2\right)^2 \;=\; -8a^2\alpha\, x - 4a^2(z^2 + \gamma^2) + \cdots,$$

which, considering z as a constant, is of the form

$$(x^2 + y^2 - \alpha)^2 \;=\; 16\,A\,(x - m);$$

that is, the section of the surface by a plane parallel to the plane of the conic is a Cartesian.

If a and b are unequal, but if we still have $\beta = 0$, the equation of the surface is

$$\left(x^2 + y^2 + z^2 - \alpha^2 - \gamma^2\right)^2 \;=\; 4a^2(x-\alpha)^2 + 4b^2y^2.$$

There are here two planes parallel to the plane of the conic, each of them meeting the surface in a pair of circles. In fact, writing $x^2 + y^2 = \rho$, and therefore also $y^2 = \rho - x^2$, putting moreover $z^2 - \alpha^2 - \gamma^2 = k$, we have

$$(\rho + k)^2 \;=\; 4a^2x^2 - 8a^2\alpha\, x + 4a^2\alpha^2 + 4b^2(\rho - x^2);$$

that is,

$$\rho^2 + 4(b^2 - a^2)x^2 + k^2 - 4a^2\alpha^2 + 8a^2\alpha\, x + (2k - 4b^2)\rho \;=\; 0,$$

or, as this may also be written,

$$\left(1,\ 4(b^2 - a^2),\ k^2 - 4a^2\alpha^2,\ 4a^2\alpha,\ k - 2b^2,\ 0 \mid \rho,\ x,\ 1\right)^2 \;=\; 0,$$

which is of the form

$$\left(a,\ b,\ c,\ f,\ g,\ 0 \mid \rho,\ x,\ 1\right)^2 \;=\; 0;$$

and the left-hand side will break up into factors, each of the form $\rho + Ax + B$ (so that, equating either factor to zero, we have $\rho + Ax + B = 0$, that is, $x^2 + y^2 + Ax + B = 0$, the equation of a circle), if only

$$a\,b\,c - a\,f^2 - b\,g^2 \;=\; 0.$$

Writing this under the form $b(ac - g^2) - af^2 = 0$, and substituting for a, b, c, f, g their values, we have

$$b = 4(b^2 - a^2), \quad ac - g^2 = k^2 - 4a^2\alpha^2 - (k - 2b^2)^2 = 4(b^2k - b^4 - a^2\alpha^2), \quad af^2 = 16a^4\alpha^2,$$

and therefore the condition is

$$(b^2 - a^2)(b^2k - b^4 - a^2\alpha^2) - a^4\alpha^2 \;=\; 0;$$

that is,

$$b^2\left\{(b^2 - a^2)(k - b^2) - a^2\alpha^2\right\} \;=\; 0.$$

If $b^2 = 0$, the surface is a pair of spheres; rejecting this factor, we have $(b^2 - a^2)(k - b^2) - a^2\alpha^2 = 0$; or putting for k its value, the condition becomes

$$(b^2 - a^2)(z^2 - \alpha^2 - \gamma^2 - b^2) - a^2\alpha^2 \;=\; 0;$$

that is, for each of the values of z given by this equation, the section by a plane parallel to the plane of the conic will be a pair of circles.

The planes in question will coincide with the plane of the conic, if only

$$(b^2 - a^2)(\alpha^2 + \gamma^2 + b^2) + a^2\alpha^2 \;=\; 0,$$

or, what is the same thing,

$$b^2\alpha^2 - (a^2 - b^2)\gamma^2 \;=\; b^2(a^2 - b^2);$$

that is, if the point $(\alpha, 0, \gamma)$ be situated on the hyperbola $y = 0$, $\dfrac{x^2}{a^2 - b^2} - \dfrac{z^2}{b^2} = 1$. The hyperbola in question and the ellipse $z = 0$, $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, are, it is clear, conics in planes at right angles to each other, having the transverse axes coincident in direction, and being such that each curve passes through the foci of the other curve; or, what is the same thing, they are a pair of focal conics of a system of confocal ellipsoids.

The surface in the case in question, viz. when the parameters a, b, α, β are connected by the equation

$$\frac{\alpha^2}{a^2 - b^2} - \frac{\gamma^2}{b^2} \;=\; 1,$$

is in fact the "Cyclide" of Dupin. It is to be noticed that we have here

$$(\alpha - a\cos\theta)^2 + b^2\sin^2\theta + \gamma^2 \;=\; \alpha^2 + \gamma^2 + b^2 - 2a\alpha\cos\theta + (a^2 - b^2)\cos^2\theta;$$

which, observing that $\alpha^2 + \gamma^2 + b^2$ is $= \dfrac{a^2\alpha^2}{a^2 - b^2}$, gives

$$(\alpha - a\cos\theta)^2 + b^2\sin^2\theta + \gamma^2 \;=\; \left(\sqrt{a^2 - b^2}\,\cos\theta - \frac{a\alpha}{\sqrt{a^2 - b^2}}\right)^2,$$

so that the radius of the variable sphere is

$$= \sqrt{a^2 - b^2}\,\cos\theta - \frac{a\alpha}{\sqrt{a^2 - b^2}}.$$

If the variable sphere, instead of passing through the point $(\alpha, 0, \gamma)$ on the hyperbola, be drawn so as to touch a sphere of radius l, having its centre at the point in question, then the radius of the variable sphere would be

$$= \sqrt{a^2 - b^2}\,\cos\theta - \frac{a\alpha}{\sqrt{a^2 - b^2}} - l,$$

which is in fact

$$= \sqrt{a^2 - b^2}\,\cos\theta - \frac{a\alpha'}{\sqrt{a^2 - b^2}}$$

if only $\alpha' = \alpha + l\sqrt{a^2 - b^2}/a$; hence if γ' be the corresponding value of γ, the variable sphere passes through the point $(\alpha', 0, \gamma')$ on the hyperbola, and the envelope is still a cyclide. The cyclide as derived from the foregoing investigation is thus the envelope of a sphere having its centre on the ellipse, and touching a fixed sphere having its centre on the hyperbola. It also appears that there are, having their centres on the hyperbola, an infinite series of spheres each touched by the spheres which have their centre on the ellipse; if, instead of one of these spheres we take any four of them, this will imply that the centre of the variable sphere is on the ellipse, and it is thus seen that the cyclide as obtained above is identical with the cyclide according to the original definition, viz. as the envelope of a sphere touching four given spheres.

Cambridge, December 5, 1864.

Paper 361: On Quartic Curves

[*From the Philosophical Magazine*, vol. XXIX. (1865), pp. 105–108.]

The expression 'an oval' is used, in regard to the plane, to denote a closed curve without nodes or cusps; and, in regard to the sphere, it is assumed moreover that the oval is a curve which is not its own opposite, and does not meet the opposite curve[1]—that is, that the oval is one of a pair of non-intersecting twin ovals. I say that every spherical curve of the fourth order (or spherical quartic) without nodes or cusps may be considered as composed of an oval or ovals lying wholly in one hemisphere (that is, not cutting or touching the bounding circle of the hemisphere), and of the opposite oval or ovals lying wholly in the opposite hemisphere; or, disregarding the opposite curves, that it consists of an oval or ovals lying wholly

[1]The notions of opposite curves, &c., are fully developed in the excellent Memoir of Möbius, "Ueber die Grundformen der Linien der dritter Ordnung," *Abh. der K. Sächs. Ges. zu Leipzig*, vol. I. (1852), to which I have elsewhere frequently referred.

in one hemisphere. And this being so, the quartic cone having its vertex at the centre of the sphere is met by a plane parallel to that of the bounding circle in a plane quartic curve consisting of an oval or ovals; and thence every plane quartic is either a finite curve consisting of an oval or ovals, or else the projection of such a curve.

Considering first the case of the plane, a line in general meets the oval in an even number of points (the number may of course be $= 0$); hence as the point of contact of a tangent reckons for two points, the tangent at any point of the oval again intersects the oval in an even number of points (this number may of course be $= 0$). The number of points of intersection by the tangent (the point of contact being always excluded) is either evenly even, and the point is then situate on a convex portion of the oval; or it is oddly even, and the point is then situate on a concave portion of the oval. Now imagine that the oval is (or is part of) a quartic curve; the number of points of intersection by the tangent is $= 0$ or else $= 2$; and there is at least one portion of the oval for which the number of intersections is $= 0$; for otherwise the oval would be concave at every point, which is impossible. Hence there is a tangent which does not meet the oval (except at the point of contact), and we may in the immediate neighbourhood of the tangent draw a line which does not meet the oval at all.

Precisely the same considerations apply to the case of an oval which is part of a spherical quartic, the tangent being of course a great circle; and the conclusion arrived at is that there exists a great circle which does not meet the oval at all; that is, the oval lies wholly in one hemisphere.

I remark that the demonstration would, as it ought to do, fail, if we attempted to apply it to an oval portion of a spherical sextic; the tangent circle meets the oval in a number of points which is $= 0$, 2, or 4; and the number cannot be for every tangent circle whatever $= 2$; but there is nothing to prevent it from being for every tangent circle whatever $= 2$ or 4. Hence we cannot, for every spherical sextic, obtain a tangent circle not meeting the oval except at the point of contact; and consequently we do not obtain in the immediate neighbourhood of the tangent a circle which does not meet the oval at all. And in fact such circle does not in every case exist; that is, the oval portion of a spherical sextic does not in every case lie in a hemisphere.

It has been shown that the oval portion of a spherical quartic lies in a hemisphere; but we have to consider the case where the quartic consists of two or more ovals. To fix the ideas, let A, A' be a pair of opposite ovals, and B, B' another pair of opposite ovals, components of the same spherical quartic. If there exists a tangent circle of A which does not meet B, then there exists in the immediate neighbourhood of the tangent circle a circle which does not meet either A or B; and we may assume that A and B lie on the same side of this circle; for if B were on the side opposite to A, then B' would be on the same side with A; and we have only, instead of B, to consider the opposite oval B'. Hence we may consider that the ovals A and B lie on the same side of the circle; that is, we have a spherical quartic consisting of or comprising the ovals A and B in the same hemisphere: the two ovals are, it is clear, external each to the other.

But every tangent of A may meet B in two points; consider the whole spherical figure, and suppose that the tangent (or say, the tangent circle) of A, A' meets the ovals B, B' in the points K, L and the opposite points K', L': then considering the tangent circle as moving round A, A' until it returns to its original position, the points K, L, K', L' are always four distinct points; and K and some one (say L) of the two points L, L' will describe the same oval, say the oval B; while the opposite points K', L' will describe the opposite oval B'. We have here the oval A included in the oval B (and of course the opposite oval A' included in the opposite oval B'). But the oval B, quâ portion of a spherical quartic, lies wholly in one hemisphere; hence the two ovals A, B lie wholly in one hemisphere. It is easy to see that there is not in this case any other portion of the spherical quartic, but that the two ovals A, B are the entire curve.

Reverting to the case where we have in one hemisphere the two ovals A, B external to each other, the spherical quartic may comprise as part of itself another oval C. The ovals A and B, quâ ovals external to each other, have a common tangent circle (a double tangent of the spherical quartic) which cannot meet the oval C (for if it did we should have six points of intersection); hence in the immediate neighbourhood thereof we have a circle not meeting any one of the ovals A, B, C. We may consider A, B, C as lying on the same side of this circle; for if B were on the opposite side to A, then B' would be on the same side; that is, we have if C be on the opposite side, then C' will be on the same side; and so if C be on the opposite side, then C' will be on the same side; that is, we have the three ovals A, B, C external to each other, and in the same hemisphere.

There may be a fourth oval, D, and it would be shown in a similar manner that we have then the four ovals A, B, C, D external to each other and in the same hemisphere. But there cannot be a fifth oval, E;

the proof is precisely the same as for the theorem *in plano*; viz. taking within each of the five ovals a point, and through these points drawing a conic, the conic would meet each oval in two points, and therefore the plane quartic in ten points, which is impossible.

Passing from the sphere to the plane, the foregoing investigation shows that every plane quartic without nodes or cusps is either a finite curve, or else the projection of a finite curve, of one of the following forms:

1. a single oval.

2. two ovals external to each other.

3. two ovals, one inside the other.

4. three ovals external to each other.

5, 6. four ovals external to each other.

The last case has been called (5, 6) for the sake of the following subdivision, viz.:

5. the four ovals are so situate as to be intersected, each in two points, by the same ellipse.

6. they are so situate as not to be intersected by any one ellipse whatever—the distinction being similar to that which exists between four points, which may be either such as to have passing through them as well ellipses as hyperbolas, or else to have passing through them hyperbolas only.

I remark that the limitation of the theorem to the case of a quartic curve without nodes or cusps is necessary, at any rate as regards the nodes. We may in fact find a quartic curve having a single node which is met by every line in at least two real points, and which is therefore not the projection of any finite curve; for if we imagine two hyperbolas so situate that each branch of the one cuts each branch of the other, then it may be seen that there exists a quartic curve approaching everywhere very nearly to the system of two hyperbolas, but having, instead of the four nodes of the system, only a single node, which is such that every line meets it in at least two points.

Cambridge, December 15, 1864.

Paper 362: Note on Lobatschewsky's Imaginary Geometry

[*From the Philosophical Magazine*, vol. XXIX. (1865), pp. 231–233.]

Writing down the equations

$$\frac{1}{\cos a'} = \cos a = \frac{\cos A + \cos B \cos C}{\sin B \sin C},$$
$$\frac{1}{\cos b'} = \cos b = \frac{\cos B + \cos C \cos A}{\sin C \sin A},$$
$$\frac{1}{\cos c'} = \cos c = \frac{\cos C + \cos A \cos B}{\sin A \sin B},$$

where A, B, C are real positive angles each $< \frac{1}{2}\pi$: first, if $A+B+C > \pi$, then a, b, c are real positive angles each $< \frac{1}{2}\pi$ (this is in fact the case of a real acute-angled spherical triangle), but a', b', c' are pure imaginaries of the form $p'i$, $q'i$, $r'i$ (where p', q', r' are real positive quantities); and secondly, if $A+B+C < \pi$, then a, b, c are pure imaginaries of the form pi, qi, ri (where p, q, r are real positive quantities), but a', b', c' are real positive angles each less than $\frac{1}{2}\pi$. Hence assuming $A+B+C < \pi$ and writing ai, bi, ci in place of a, b, c, the system is

$$\frac{1}{\cos a'} = \cos ai = \frac{\cos A + \cos B \cos C}{\sin B \sin C},$$
$$\frac{1}{\cos b'} = \cos bi = \frac{\cos B + \cos C \cos A}{\sin C \sin A},$$
$$\frac{1}{\cos c'} = \cos ci = \frac{\cos C + \cos A \cos B}{\sin A \sin B},$$

which equations (if only we write therein $\frac{1}{2}\pi - a'$, $\frac{1}{2}\pi - b'$, $\frac{1}{2}\pi - c'$ in place of a', b', c' respectively) are in fact the equations given under a less symmetrical form in the curious paper "Géométrie Imaginaire" by N. Lobatschewsky, Rector of the University of Kasan, *Crelle*, vol. XVII. (1837), pp. 295–320. The view taken of them by the author is hard to be understood. He mentions that in a paper published five years previously in a scientific journal at Kasan, after developing a new theory of parallels, he had endeavoured to prove that it is only experience which obliges us to assume that in a rectilinear triangle the sum of the angles is equal to two right angles, and that a geometry may exist, if not in nature at least in analysis, on the hypothesis that the sum of the angles is less than two right angles; and he accordingly attempts to establish such a geometry, viz. a, b, c being the sides of a rectilinear triangle, wherein the sum of the angles $A+B+C$ is $<\pi$, and the angles a', b', c' being calculated from the sides by the formulae

$$\frac{1}{\cos a'} = \cos ai, \quad \frac{1}{\cos b'} = \cos bi, \quad \frac{1}{\cos c'} = \cos ci.$$

(I have, as mentioned above, replaced Lobatschewsky's a', b', c' by their complements): the relation between the angles A, B, C and the subsidiary quantities a', b', c' which replace the sides, is given by the formulae

$$\begin{aligned}
\frac{1}{\cos a'} &= \frac{\cos A + \cos B \cos C}{\sin B \sin C}, \\
\frac{1}{\cos b'} &= \frac{\cos B + \cos C \cos A}{\sin C \sin A}, \\
\frac{1}{\cos c'} &= \frac{\cos C + \cos A \cos B}{\sin A \sin B}.
\end{aligned}$$

I do not understand this; but it would be very interesting to find a real geometrical interpretation of the last-mentioned system of equations, which (if only A, B, C are positive real quantities such that $A+B+C<\pi$; for the condition, A, B, C each $<\frac{1}{2}\pi$, may be omitted) contains only *real* quantities A, B, C, a', b', c'; and is a system correlative to the equations of ordinary Spherical Trigonometry.

It is hardly necessary to remark that the equation

$$\frac{1}{\cos a} = \cos ai$$

is Jacobi's imaginary transformation in the Theory of Elliptic Functions. See, as to this, my paper "On the Transcendent $\text{gd}\,.u = -i \log \tan(\frac{1}{4}\pi + \frac{1}{2}ui)$," *Phil. Mag.*, vol. XXIV. (1862), pp. 19–22, [320].

Cambridge, January 21, 1865.

Paper 363: On the Theory of the Evolute

[*From the Philosophical Magazine*, vol. XXIX. (1865), pp. 344–350.]

According to the generalized notion of geometrical magnitude, two lines are said to be at right angles to each other when they are harmonics in regard to a certain conic called the Absolute; this being so, the normal at any point of a curve is the line at right angles to the tangent, and the Evolute is the envelope of the normals.

Let the equation of the absolute be

$$\Omega = (a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

and suppose, as usual, that the inverse coefficients are (A, B, C, F, G, H). Consider a given curve $U = (*)(x,y,z)^m = 0$, and suppose, for shortness, that the first differential coefficients of U are denoted by L, M, N. Then we have to find the equation of the normal at the point (x, y, z) of the curve $U = 0$.

The condition that any two lines are harmonics in regard to the absolute, is equivalent to this, viz. each line passes through the pole of the other line in regard to the absolute. Hence the normal at the point (x, y, z) is the line joining this point with the pole of the tangent. Now, taking (X, Y, Z) as current coordinates, the equation of the tangent is

$$LX + MY + NZ = 0,$$

the coordinates of the pole of the tangent are therefore

$$(A,\ H,\ G \mid L,\ M,\ N)\ :\ (H,\ B,\ F \mid L,\ M,\ N)\ :\ (G,\ F,\ C \mid L,\ M,\ N),$$

and the equation of the normal is

$$\begin{vmatrix} X & Y & Z \\ x & y & z \\ (A,H,G \mid L,M,N) & (H,B,F \mid L,M,N) & (G,F,C \mid L,M,N) \end{vmatrix} = 0.$$

The formula in this form will be convenient in the sequel; but there is no real loss of generality in taking the equation of the absolute to be $x^2+y^2+z^2=0$; the values of (A,B,C,F,G,H) are then $(1,1,1,0,0,0)$, and the formula becomes

$$\begin{vmatrix} X & Y & Z \\ x & y & z \\ L & M & N \end{vmatrix} = 0;$$

where it will be remembered that (L,M,N) denote the derived functions $(\partial_x U,\ \partial_y U,\ \partial_z U)$.

The evolute is therefore the envelope of the line represented by the foregoing equation, say the equation $\Omega=0$, considering therein (x,y,z) as variable parameters connected by the equation $U=0$.

As an example, let it be required to find the evolute of a conic; since the axes are arbitrary, we may without loss of generality assume that the equation of the conic is $xz-y^2=0$. The values of (L,M,N) here are $(z,-2y,x)$. Moreover the equation is satisfied by writing therein $x:y:z=1:\theta:\theta^2$; the values of (L,M,N) then become $(\theta^2,-2\theta,1)$ and the equation is

$$\begin{vmatrix} 1 & \theta & \theta^2 \\ X & Y & Z \\ (A,H,G \mid \theta^2,-2\theta,1) & (H,B,F \mid \theta^2,-2\theta,1) & (G,F,C \mid \theta^2,-2\theta,1) \end{vmatrix} = 0,$$

or, developing, this is

$$\begin{aligned} &X\big(G\theta^4-2F\theta^3+C\theta^2-\cdots\big)\\ &\quad+(-H\theta^4+2B\theta^3-F\theta^2+\cdots)\\ &+Y\big(A\theta^4-2H\theta^3+G\theta^2+2F\theta-\cdots\big)\\ &+Z\big(H\theta^4-2B\theta^3+F\theta^2\\ &\quad-A\theta^4+2H\theta^3-G\theta^2\big)\ =\ 0,\end{aligned}$$

which I leave in this form in order to show the origin of the different terms, and in particular in order to exhibit the destruction of the term θ^4 in the coefficient of Y. But the equation is, it will be observed, a quartic equation in θ, with coefficients which are linear functions of the current coordinates (X,Y,Z).

The equation shows at once that the evolute is of the class 4; in fact treating the coordinates (X,Y,Z) as given quantities, we have for the determination of θ an equation of the order 4, that is, the number of normals through a given point (X,Y,Z), or, what is the same thing, the class of the evolute, is $=4$.

The equation of the evolute is obtained by equating to zero the discriminant of the foregoing quartic function of θ; the order of the evolute is thus $=6$. There are no inflexions, and the diminution of the order from $4\cdot 3$, $=12$, to 6 is caused by three double tangents.

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